

|                                   |                                                                                                                                                                 |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Name</b>               | RTDIP Timeseries Forecasting                                                                                                                                    |
| <b>Online team meeting</b>        | <a href="https://fau.zoom-x.de/j/65502405196?pwd=8H73lyixQfqKrnO8Eb47kawnuKCnGp.1">https://fau.zoom-x.de/j/65502405196?pwd=8H73lyixQfqKrnO8Eb47kawnuKCnGp.1</a> |
| <b>Production system (if any)</b> |                                                                                                                                                                 |
| <b>Test system (if any)</b>       |                                                                                                                                                                 |
| <b>GitHub repository</b>          | <a href="https://github.com/amosproj/amos2025ws03-rtdip-timeseries-forecasting">https://github.com/amosproj/amos2025ws03-rtdip-timeseries-forecasting</a>       |
| <b>GitHub feature board</b>       | <a href="https://github.com/orgs/amosproj/projects/91/">https://github.com/orgs/amosproj/projects/91/</a>                                                       |
| <b>GitHub imp-squared backlog</b> | <a href="https://github.com/orgs/amosproj/projects/96">https://github.com/orgs/amosproj/projects/96</a>                                                         |
| <b>Team T-shirt (white)</b>       | <a href="https://www.shirtinator.de/s/QWcDXTKGS72ISnrRNZq1hg">https://www.shirtinator.de/s/QWcDXTKGS72ISnrRNZq1hg</a>                                           |
| <b>Team T-shirt (black)</b>       | <a href="https://www.shirtinator.de/s/0yx3duRjSzW-5hP32KmZ1w">https://www.shirtinator.de/s/0yx3duRjSzW-5hP32KmZ1w</a>                                           |
| <b>Additional materials</b>       | <a href="https://discord.gg/KJXGmjcs">https://discord.gg/KJXGmjcs</a>                                                                                           |
| <b>Team mailing list</b>          | oss-amos-projX@lists.fau.de                                                                                                                                     |
| <b>Happines Index App</b>         | <a href="https://happy-amos.appspot.com/Courses">https://happy-amos.appspot.com/Courses</a>                                                                     |
| <b>Planning Poker</b>             | <a href="https://planningpokeronline.com/DMhnF5cOAK9ffF5jsRf2/">https://planningpokeronline.com/DMhnF5cOAK9ffF5jsRf2/</a>                                       |

[illegible]

| #                                                                                                                                                                       | Meeting Day | Product Owner    |                  | Software Developer | Release Manager | Scrum Master | Comment       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------|------------------|--------------------|-----------------|--------------|---------------|
|                                                                                                                                                                         |             | Review           | Planning         |                    |                 |              |               |
| 1                                                                                                                                                                       | 2025-10-15  | Patrick Meusling | Hafidz Arifin    | Everyone else      | N/A             | Luca         |               |
| 2                                                                                                                                                                       | 2025-10-22  | Hafidz Arifin    | Patrick Meusling | Everyone else      | Christoph       | Luca         |               |
| 3                                                                                                                                                                       | 2025-10-29  | Patrick Meusling | Hafidz Arifin    | Everyone else      | Mehdi Khabouze  | Luca         |               |
| 4                                                                                                                                                                       | 2025-11-05  | Hafidz Arifin    | Patrick Meusling | Everyone else      | Rene Jokiel     | Luca         |               |
| 5                                                                                                                                                                       | 2025-11-12  | Patrick Meusling | Hafidz Arifin    | Everyone else      | Abdul Haseeb    | Luca         |               |
| 6                                                                                                                                                                       | 2025-11-19  | Hafidz Arifin    | Patrick Meusling | Everyone else      | Hannes Pohnke   | Luca         |               |
| 7                                                                                                                                                                       | 2025-11-26  | Patrick Meusling | Hafidz Arifin    | Everyone else      | Simon Selbig    | Luca         | Mid-term due  |
| 8                                                                                                                                                                       | 2025-12-03  | Hafidz Arifin    | Patrick Meusling | Everyone else      | Rene Jokiel     | Luca         |               |
| 9                                                                                                                                                                       | 2025-12-10  | Patrick Meusling | Hafidz Arifin    | Everyone else      | Mehdi Khabouze  | Luca         |               |
| 10                                                                                                                                                                      | 2025-12-17  | Hafidz Arifin    | Patrick Meusling | Everyone else      | Christoph Huy   | Luca         |               |
| 11                                                                                                                                                                      | 2026-01-07  | Patrick Meusling | Hafidz Arifin    | Everyone else      | Hannes Pohnke   | Luca         |               |
| 12                                                                                                                                                                      | 2026-01-14  | Hafidz Arifin    | Patrick Meusling | Everyone else      | Simon Selbig    | Luca         |               |
| 13                                                                                                                                                                      | 2026-01-21  | Patrick Meusling | Hafidz Arifin    | Everyone else      | Mehdi Khabouze  | Luca         |               |
| 14                                                                                                                                                                      | 2026-01-28  | Hafidz Arifin    | Patrick Meusling | Everyone else      | Hannes Pohnke   | Luca         |               |
| 15                                                                                                                                                                      | 2026-02-04  | Patrick Meusling | Hafidz Arifin    | Everyone else      | Christoph Huy   | Luca         | Demo day!     |
|                                                                                                                                                                         | 2026-02-11  | Hafidz Arifin    | Patrick Meusling | Everyone else      |                 | Luca         | Retrospective |
| Product owners, software developers, and Scrum Master are set and ideally don't change over time; the critical part is the Release Manager role you need to define here |             |                  |                  |                    |                 |              |               |
|                                                                                                                                                                         |             |                  |                  |                    |                 |              |               |

|                                |                                                                                           |
|--------------------------------|-------------------------------------------------------------------------------------------|
| <b>Goals</b>                   | Aquire new skills                                                                         |
|                                | Produce a functioning and valuable product (a 1.0)                                        |
| <b>Meeting norms</b>           | We respect the opinions of others and we show up on time                                  |
|                                | Summary of Partner Meeting required                                                       |
| <b>Working norms</b>           | Clean, testable code with clear commit messages, code comments                            |
|                                | We respect other people's work                                                            |
| <b>Coordination norms</b>      | Seperate branches for issues, second person review                                        |
|                                | We balance workload among the team                                                        |
| <b>Communication norms</b>     | Daily check the corresponding channels (discord)                                          |
|                                | We communicate constructively, make sure to communicate possible problems (absence, etc.) |
| <b>Consideration norms</b>     | We discuss issues openly                                                                  |
|                                | We vote in case we can't reach a consensus                                                |
| <b>Cont. improvement norms</b> | We encourage critique and improvement efforts                                             |
|                                | Encourage reaching out for help (use the strength/skills of others)                       |
| <b>Rewards</b>                 | We treat ourselves to a sweet of choice for good work                                     |
|                                |                                                                                           |
| <b>Sanctions</b>               | Push Ups (amount decided on the case)                                                     |
|                                |                                                                                           |
| <b>Signatures</b>              |                                                                                           |
|                                |                                                                                           |
| Scrum Master                   | Luca Böhm                                                                                 |
| Product owner                  | Patrick Meusling                                                                          |
| Product owner                  | Hafidz Arifin                                                                             |
| Software developer             | Abdul Haseeb                                                                              |
| Software developer             | Rene Jokiel                                                                               |
| Software developer             | Simon Selbig                                                                              |
| Software developer             | Hannes Pohnke                                                                             |
| Software developer             | Christoph Huy                                                                             |
| Software developer             | Mehdi Khabouze                                                                            |

| Product Vision                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Project Mission                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To empower organizations with open-source, scalable, and transparent forecasting capabilities that enable data-driven decision-making across industries.</p> <p>By extending RTDIP with advanced forecasting and anomaly detection components as well as sector specific datasets, we envision a future where time series insights such as trends, seasonality, and predictive patterns are seamlessly integrated into real-time data pipelines.</p> <p>This will allow businesses to understand historical data, anticipate future behavior, optimize operations, and improve efficiency using accessible, community-driven, and production-ready forecasting tools.</p> | <p>To design, develop, and contribute open-source forecasting components for the RTDIP platform that enable trend analysis, anomaly detection, and predictive modeling on time series data.</p> <p>Our mission within this project is to research and implement forecasting techniques using Python and Apache Spark, validate and enrich them with real-world datasets, and ensure they meet RTDIP's modular, tested, and well-documented standards.</p> <p>By doing so, we will enhance RTDIP's functionality and provide the open-source community and industry users such as Shell with reliable, scalable forecasting capabilities that integrate seamlessly into existing data pipelines.</p> |

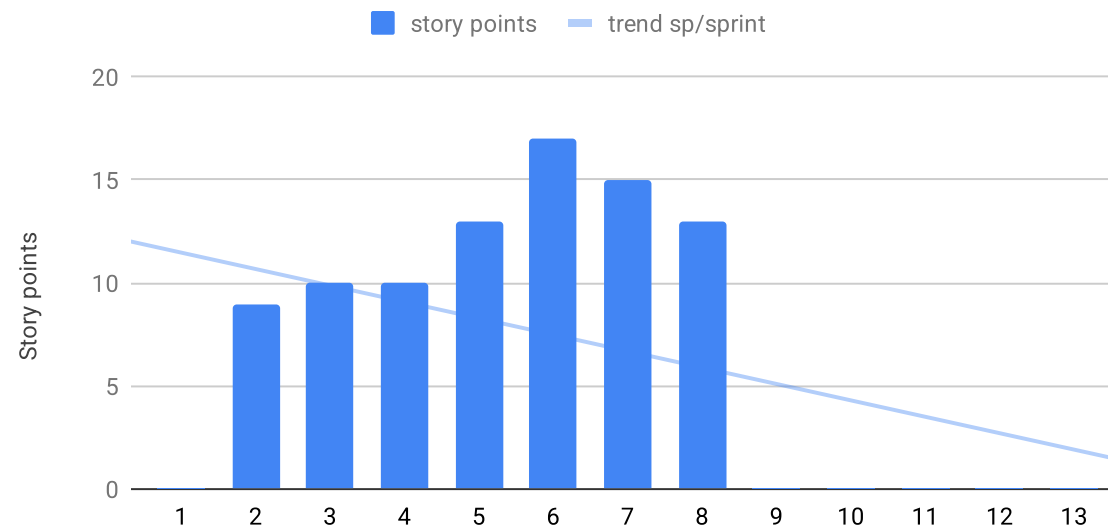
| Term                                         | Definition                                                                                                                                                                        |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RTDIP (Real-Time Data Ingestion Platform)    | An open-source project under the Linux Foundation designed to simplify the ingestion, transformation, and storage of data from various sources using scalable cloud technologies. |
| Forecasting                                  | The process of predicting future values or trends based on historical time series data.                                                                                           |
| Time Series Data                             | Data collected over time at regular intervals, often used to analyze patterns such as trends and seasonality.                                                                     |
| Apache Spark                                 | An open-source distributed computing system that provides fast data processing for large-scale datasets.                                                                          |
| PySpark                                      | The Python API for Apache Spark, allowing developers to write Spark applications using Python.                                                                                    |
| Delta Lake                                   | A storage layer that brings reliability, consistency, and performance to data lakes by supporting ACID transactions and schema enforcement.                                       |
| Anomaly Detection                            | The identification of unusual patterns or outliers in data that do not conform to expected behavior.                                                                              |
| Predictive Modeling                          | A statistical or machine learning approach that uses historical data to predict future outcomes.                                                                                  |
| Time Series Decomposition                    | The process of breaking down a time series into its core components such as trend, seasonality, and residuals.                                                                    |
| ETL (Extract, Transform, Load)               | A data pipeline process that extracts data from sources, transforms it into a suitable format, and loads it into a storage system.                                                |
| Open Source                                  | Software that is freely available to use, modify, and distribute, typically developed collaboratively by a community.                                                             |
| Linux Foundation                             | A nonprofit organization that supports open-source software projects and fosters collaboration across industries.                                                                 |
| EasyCLA (Easy Contributor License Agreement) | A system used by the Linux Foundation to manage contributor license agreements, ensuring legal compliance for open-source contributions.                                          |
| Modular Architecture                         | A software design principle that divides a system into separate, interchangeable components that can be developed and maintained independently.                                   |
| Unit Testing                                 | The practice of testing individual units or components of software to ensure they work as intended.                                                                               |
| Documentation                                | Written descriptions and guides that explain the design, usage, and functionality of a software system.                                                                           |
| Data Pipeline                                | A set of processes that move, transform, and store data from one system to another in a structured and automated way.                                                             |
| Scalability                                  | The ability of a system to handle increasing amounts of work or data by adding resources.                                                                                         |
| Contribution Guidelines                      | A set of rules and best practices that contributors must follow to ensure consistency and quality in open-source projects.                                                        |

| Sprint # | Sprint goal                                                                                            |
|----------|--------------------------------------------------------------------------------------------------------|
| 1        | None                                                                                                   |
| 2        | None                                                                                                   |
| 3        | None                                                                                                   |
| 4        | Optional                                                                                               |
| 5        | Establish a working foundation for the forecasting pipeline                                            |
| 6        | Integrate the pipeline into the existing software                                                      |
| 7        | Advance the pipeline                                                                                   |
| 8        | Add more Datasets to the pipeline                                                                      |
| 9        | Testing and Analysing the new Datasets                                                                 |
| 10       | Adding new and different models to the pipeline                                                        |
| 11       | Testing and Analysing the new Models                                                                   |
| 12       | Finalize the Product                                                                                   |
| 13       | Finalize for Demo day                                                                                  |
| 14       | Merging the product with the partner, coordinating the merge (they said this might take a bit of time) |
| 15       | Backup Sprint                                                                                          |
|          |                                                                                                        |
|          |                                                                                                        |
|          |                                                                                                        |

| Sprint # | Story Points Realized                                                      |
|----------|----------------------------------------------------------------------------|
| 1        | 0                                                                          |
| 2        | 9                                                                          |
| 3        | 10                                                                         |
| 4        | 10                                                                         |
| 5        | 13                                                                         |
| 6        | 17                                                                         |
| 7        | 15                                                                         |
| 8        | 13                                                                         |
| 9        | 0                                                                          |
| 10       | 0                                                                          |
| 11       | 0                                                                          |
| 12       | 0                                                                          |
| 13       | 0                                                                          |
| 14       | 0                                                                          |
|          |                                                                            |
|          |                                                                            |
|          | PLEASE CREATE THE VELOCITY CHART ON A NEW TAB USING THE DATA FROM THIS TAB |
|          |                                                                            |

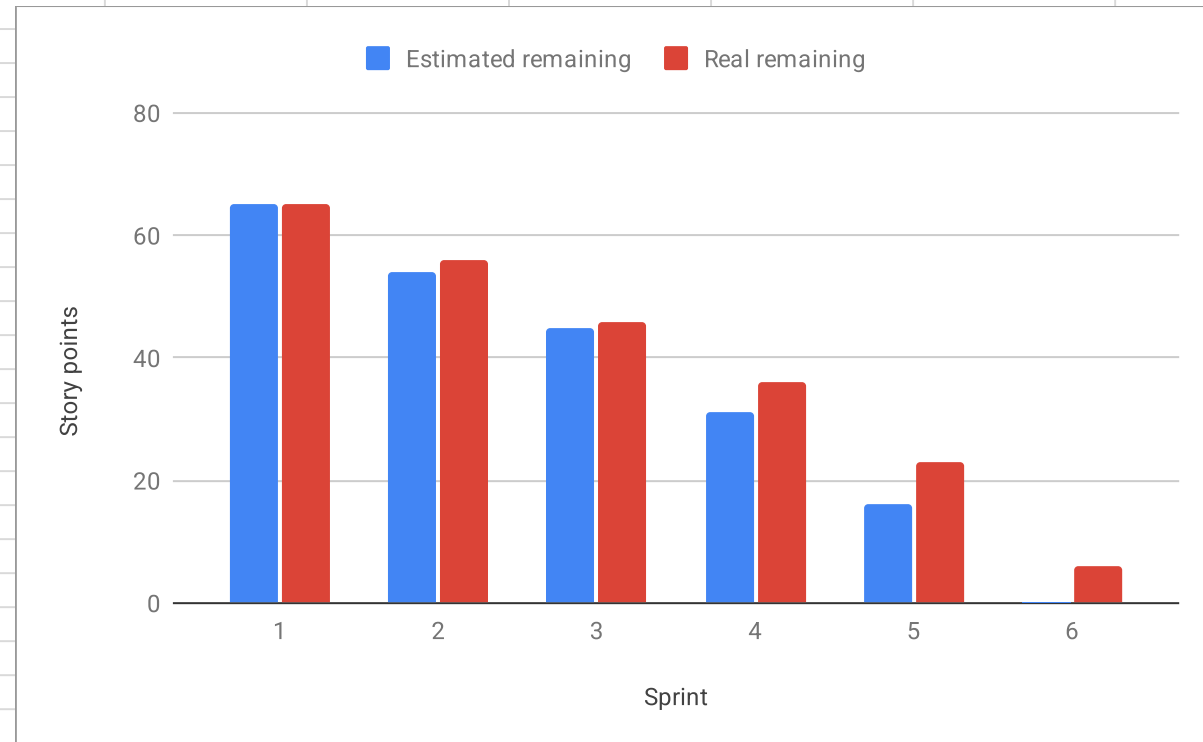


Team 3 Velocity Chart



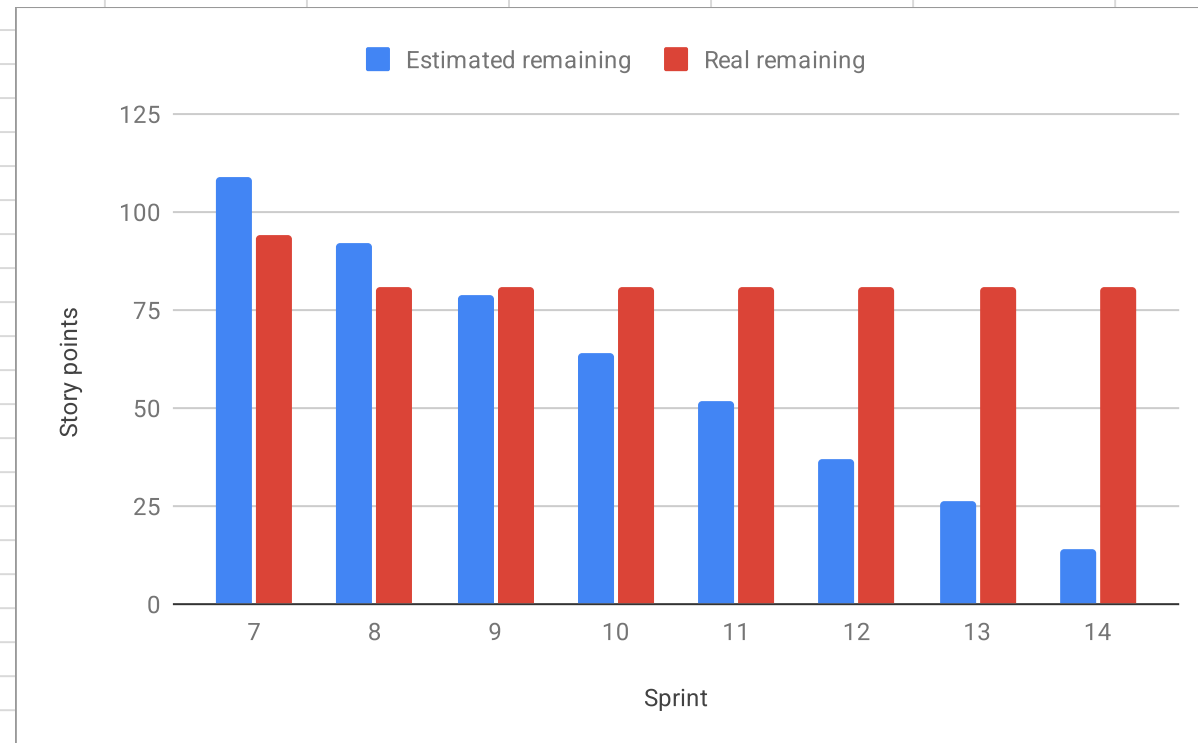
| Sprint   | Goal                                                        | Feature Name                                           | Est. size | Est. remaining | Real size    | Real remaining |
|----------|-------------------------------------------------------------|--------------------------------------------------------|-----------|----------------|--------------|----------------|
| Release  |                                                             |                                                        |           |                |              |                |
| Total    |                                                             |                                                        | 65        | 65             |              |                |
| Sprints  |                                                             |                                                        |           |                |              |                |
| 1        | None                                                        |                                                        | 0         | 65             | 0            | 65             |
| 2        | None                                                        |                                                        | 11        | 54             | 9            | 56             |
| 3        | None                                                        |                                                        | 9         | 45             | 10           | 46             |
| 4        | Optional                                                    |                                                        | 14        | 31             | 10           | 36             |
| 5        | Establish a working foundation for the forecasting pipeline |                                                        | 15        | 16             | 13           | 23             |
| 6        | Integrate the pipeline into the existing software           |                                                        | 16        | 0              | 17           | 6              |
| Features |                                                             |                                                        |           |                |              |                |
| 1        | None                                                        |                                                        |           |                |              |                |
| 2        | None                                                        |                                                        |           |                |              |                |
|          |                                                             | Run the software on your own machine                   | 1         |                | 1            |                |
|          |                                                             | Understanding the rtdip software                       | 2         |                | 2            |                |
|          |                                                             | Software Architecture                                  | 2         |                | 2            |                |
|          |                                                             | Find open source time series datasets                  | 2         |                | 2            |                |
|          |                                                             | Initialize software bill of materials                  | 2         |                | 2            |                |
|          |                                                             | Research about time series forecasting fundamentals    | 2         |                | NotCompleted |                |
| 3        | None                                                        |                                                        |           |                |              |                |
|          |                                                             | Research Possible Models Useful for Shell Dataset      | 2         |                | 2            |                |
|          |                                                             | Update our Forked Repository with Latest RTDIP Release | 1         |                | 1            |                |
|          |                                                             | Prepare and Demo Project Build Process                 | 1         |                | 2            |                |
|          |                                                             | Perform Data Exploration on Shell Dataset              | 3         |                | 3            |                |
|          |                                                             | Research about time series forecasting fundamentals    | 2         |                | 2            |                |
| 4        | Optional                                                    |                                                        |           |                |              |                |
|          |                                                             | Getting to Know HPC of AMOS                            | 2         |                | NotCompleted |                |
|          |                                                             | Perform Data Exploration on Opensource Dataset 1       | 2         |                | 3            |                |
|          |                                                             | Preprocess Shell Dataset                               | 3         |                | 3            |                |
|          |                                                             | Implementing a basic Model pipeline                    | 2         |                | 2            |                |
|          |                                                             | Research Possible Models Useful fo Opensource Dataset  | 2         |                | NotCompleted |                |
|          |                                                             | CI/CD Pipeline fail bug                                | 3         |                | 2            |                |
| 5        | Establish a working foundation for the forecasting pipeline |                                                        |           |                |              |                |
|          |                                                             | Choosing Models, Datasets, Error Measure Metrics       | 2         |                | 2            |                |
|          |                                                             | Create Build Process Video                             | 2         |                | 2            |                |
|          |                                                             | Model Training and Evaluation with Shell Dataset       | 3         |                | 3            |                |
|          |                                                             | Visualization of Results                               | 2         |                | 1            |                |
|          |                                                             | Preprocess Opensource Dataset 1                        | 3         |                | 3            |                |
|          |                                                             | PO Homework                                            | 3         |                | 2            |                |

| Sprint | Goal                                              | Feature Name                                                               | Est. size | Est. remaining | Real size    | Real remaining |
|--------|---------------------------------------------------|----------------------------------------------------------------------------|-----------|----------------|--------------|----------------|
| 6      | Integrate the pipeline into the existing software |                                                                            |           |                |              |                |
|        |                                                   | Model Deployment Pipeline into RTDIP                                       | 3         |                | 3            |                |
|        |                                                   | User/Design/Build Documentation                                            | 2         |                | 2            |                |
|        |                                                   | Repository Structure Refactor                                              | 2         |                | 2            |                |
|        |                                                   | Update Software Architecture                                               | 2         |                | 2            |                |
|        |                                                   | Set Up Unified Development Environment                                     | 3         |                | 5            |                |
|        |                                                   | Collect and Document Customer Feedback on Project Progress                 | 1         |                | NotCompleted |                |
|        |                                                   | Preprocess Opensource SCADA                                                | 3         |                | 3            |                |
|        |                                                   |                                                                            |           |                |              |                |
|        |                                                   |                                                                            |           |                |              |                |
|        |                                                   |                                                                            |           |                |              |                |
|        |                                                   |                                                                            |           |                |              |                |
|        |                                                   |                                                                            |           |                |              |                |
|        |                                                   |                                                                            |           |                |              |                |
|        |                                                   |                                                                            |           |                |              |                |
|        |                                                   |                                                                            |           |                |              |                |
|        |                                                   |                                                                            |           |                |              |                |
|        |                                                   |                                                                            |           |                |              |                |
|        |                                                   |                                                                            |           |                |              |                |
|        |                                                   | PLEASE CREATE THE BURNDOWN CHART ON A NEW TAB USING THE DATA FROM THIS TAB |           |                |              |                |



| Sprint   | Goal                                                   | Feature Name                                                        | Est. size | Est. remaining | Real size     | Real remaining |
|----------|--------------------------------------------------------|---------------------------------------------------------------------|-----------|----------------|---------------|----------------|
| Release  |                                                        |                                                                     |           |                |               |                |
| Total    |                                                        |                                                                     | 109       | 109            |               |                |
| Sprints  |                                                        |                                                                     |           |                |               |                |
| 7        | Advance the pipeline                                   |                                                                     | 17        | 109            | 15            | 94             |
| 8        | Add more Datasets to the pipeline                      |                                                                     | 13        | 92             | 13            | 81             |
| 9        | Different Methods to Improve Forecast Accuracy         |                                                                     | 15        | 79             | 0             | 81             |
| 10       | Customer Feedback and Adjustments                      |                                                                     | 12        | 64             | 0             | 81             |
| 11       | Improving Forecast Accuracy                            |                                                                     | 15        | 52             | 0             | 81             |
| 12       | Smaller features for more/different preprocessing etc. |                                                                     | 11        | 37             | 0             | 81             |
| 13       | Demo Day Preparation                                   |                                                                     | 12        | 26             | 0             | 81             |
| 14       | Finish the Project                                     |                                                                     | 14        | 14             | 0             | 81             |
| Features |                                                        |                                                                     |           |                |               |                |
| 7        | Advance the pipeline                                   |                                                                     |           |                |               |                |
|          |                                                        | Collect and Document Customer Feedback on Project Progress          | 1         |                | 1             |                |
|          |                                                        | Adjust and Fix CI/CD Pipeline                                       | 3         |                | Not completed |                |
|          |                                                        | Model Optimization and Improvement                                  | 3         |                | 3             |                |
|          |                                                        | Implementation of Time Series Decomposition                         | 3         |                | 3             |                |
|          |                                                        | Implementation of Anomaly Detection                                 | 5         |                | 5             |                |
|          |                                                        | Model Training and Evaluation of Dataset SCADA                      | 2         |                | 3             |                |
| 8        | Add more Datasets to the pipeline                      |                                                                     |           |                |               |                |
|          |                                                        | Perform Data Exploration additional Opensource Dataset              | 3         |                | 3             |                |
|          |                                                        | Implement RTDIP Ingestion                                           | 2         |                | 2             |                |
|          |                                                        | Restructure and Refactor Wiki                                       | 2         |                | 2             |                |
|          |                                                        | Standardize Visualization                                           | 3         |                | 3             |                |
|          |                                                        | Model Optimization and Improvement for Timeseries Forecasting       | 3         |                | 3             |                |
| 9        | Different Methods to Improve Forecast Accuracy         |                                                                     |           |                |               |                |
|          |                                                        | Optimization and Improvement of Anomaly Detection Approach          | 3         |                |               |                |
|          |                                                        | Timeseries Forecasting Improvement with Other Models (Person 1)     | 3         |                |               |                |
|          |                                                        | Integrate Timeseries Forecasting Results Visualization into the SDK | 3         |                |               |                |
|          |                                                        | Apply Timeseries Decomposition on shell Dataset                     | 3         |                |               |                |
|          |                                                        | Timeseries Forecasting Improvement with Other Models (Person 2)     | 3         |                |               |                |
| 10       | Customer Feedback and Adjustments                      |                                                                     |           |                |               |                |
|          |                                                        | Based on Customer Feedback add changes to preprocessing             | 3         |                |               |                |
|          |                                                        | Based on Customer Feedback add changes to feature engineering       | 3         |                |               |                |
|          |                                                        | Based on Customer Feedback add changes to Visualization             | 2         |                |               |                |
|          |                                                        | Based on Customer Feedback add suggested models                     | 2         |                |               |                |
|          |                                                        | Based on Customer Feedback change library versions                  | 2         |                |               |                |
|          |                                                        |                                                                     |           |                |               |                |
|          |                                                        |                                                                     |           |                |               |                |

| Sprint | Goal                        | Feature Name                                                               | Est. size | Est. remaining | Real size | Real remaining |
|--------|-----------------------------|----------------------------------------------------------------------------|-----------|----------------|-----------|----------------|
| 11     | Improving Forecast Accuracy |                                                                            |           |                |           |                |
|        |                             | Testing our own and new model approaches                                   | 3         |                |           |                |
|        |                             | Timeseries Forecasting Improvement by Combining FEs from Best Models       | 3         |                |           |                |
|        |                             | Timeseries Forecasting Improvement with Exogenous Variables                | 3         |                |           |                |
|        |                             | Timeseries Forecasting Improvement with Other Models                       | 3         |                |           |                |
|        |                             | Timeseries Forecasting Improvement by Trying Different Versions of Models  | 3         |                |           |                |
| 12     | Finish the Product          |                                                                            |           |                |           |                |
|        |                             | Adjusting our new approaches based on Customer Feedback                    | 3         |                |           |                |
|        |                             | Final Refactoring                                                          | 2         |                |           |                |
|        |                             | Testing our own and new preprocessing approaches                           | 3         |                |           |                |
|        |                             | Testing our own and new postprocessing approaches                          | 3         |                |           |                |
|        |                             |                                                                            |           |                |           |                |
|        |                             |                                                                            |           |                |           |                |
| 13     | Demo Day Preparation        |                                                                            |           |                |           |                |
|        |                             | Create demo day slide                                                      | 3         |                |           |                |
|        |                             | Create demo video                                                          | 3         |                |           |                |
|        |                             | Finalize user, (technical) design, and build/deploy documentation          | 3         |                |           |                |
|        |                             | Final Refactoring and Cleaning Up Repository                               | 3         |                |           |                |
|        |                             |                                                                            |           |                |           |                |
|        |                             |                                                                            |           |                |           |                |
| 14     | Finish the Project          |                                                                            |           |                |           |                |
|        |                             | Create Project Report                                                      | 3         |                |           |                |
|        |                             | Tasks related to Industry Partner                                          | 5         |                |           |                |
|        |                             | Merging the code with main repo of partner                                 | 3         |                |           |                |
|        |                             | Final Bug Fixes                                                            | 3         |                |           |                |
|        |                             |                                                                            |           |                |           |                |
|        |                             |                                                                            |           |                |           |                |
|        |                             |                                                                            |           |                |           |                |
|        |                             |                                                                            |           |                |           |                |
|        |                             |                                                                            |           |                |           |                |
|        |                             |                                                                            |           |                |           |                |
|        |                             | PLEASE CREATE THE BURNDOWN CHART ON A NEW TAB USING THE DATA FROM THIS TAB |           |                |           |                |



| # | Feature Definition of Done                                                                                                           | Sprint Release Definition of Done                                                         | Project Release Definition of Done                                                                                   |
|---|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
|   | Acceptance criteria are met.                                                                                                         |                                                                                           |                                                                                                                      |
|   | Work are pushed to the Github repository.                                                                                            |                                                                                           |                                                                                                                      |
|   | Create a branch for each backlog items (coding)                                                                                      |                                                                                           |                                                                                                                      |
|   | A pull request is created for each related branch.                                                                                   |                                                                                           |                                                                                                                      |
|   | The work products in the pull requests are reviewed.                                                                                 |                                                                                           |                                                                                                                      |
|   | The corresponding branches are merged and closed.                                                                                    |                                                                                           |                                                                                                                      |
|   | The bill of materials section of the planning documents is updated.                                                                  |                                                                                           |                                                                                                                      |
|   | Work needs to be documented in the corresponding wiki section                                                                        |                                                                                           |                                                                                                                      |
|   | For new features unit test have to be written.                                                                                       |                                                                                           |                                                                                                                      |
|   | Update our forked repository with the latest RTDIP release.                                                                          |                                                                                           |                                                                                                                      |
|   | If the task involves coding, the implementation is integrated into the RTDIP framework and verified to function correctly within it. |                                                                                           |                                                                                                                      |
|   | All unit tests must pass successfully in the continuous integration (CI) pipeline.                                                   |                                                                                           |                                                                                                                      |
|   | All assignees must communicate and coordinate with each other to complete the task.                                                  |                                                                                           |                                                                                                                      |
|   |                                                                                                                                      | Release candidates with a working and meaningful update to the previous sprint is tagged. |                                                                                                                      |
|   |                                                                                                                                      | Previously established features must continue to work.                                    |                                                                                                                      |
|   |                                                                                                                                      |                                                                                           | The RTDIP forecasting components can be successfully built and deployed within the RTDIP environment.                |
|   |                                                                                                                                      |                                                                                           | All automated tests and validation checks for the developed components pass without errors.                          |
|   |                                                                                                                                      |                                                                                           | The implemented forecasting features have been tested end-to-end and verified through a basic user workflow.         |
|   |                                                                                                                                      |                                                                                           | Developer documentation is complete, clearly describing the architecture, setup, and contribution process for RTDIP. |
|   |                                                                                                                                      |                                                                                           | User documentation (usage examples, configuration steps, expected outputs) is finalized and up to date.              |
|   |                                                                                                                                      |                                                                                           | The final release has been reviewed and approved by all project team members and the industry partner (Shell).       |



[illegible]

[illegible]

| Last Name                                                                  | First Name | Value |  | #DIV/0! | #DIV/0!          |  |  |
|----------------------------------------------------------------------------|------------|-------|--|---------|------------------|--|--|
| Böhm                                                                       | Luca       |       |  |         |                  |  |  |
| Meusling                                                                   | Patrick    |       |  |         |                  |  |  |
| Arifin                                                                     | Hafidz     |       |  |         |                  |  |  |
| Huy                                                                        | Christoph  |       |  |         |                  |  |  |
| Selbig                                                                     | Simon      |       |  | 0       | No size          |  |  |
| Haseeb                                                                     | Abdul      |       |  | 1       | Trivial size     |  |  |
| Pohnke                                                                     | Hannes     |       |  | 2       | Small size       |  |  |
| Jokiel                                                                     | Rene       |       |  | 3       | Medium size      |  |  |
| Khabouze                                                                   | Mehdi      |       |  | 5       | Large size       |  |  |
|                                                                            |            |       |  | 8       | Very large size  |  |  |
|                                                                            |            |       |  | 13      | Too large (size) |  |  |
|                                                                            |            |       |  |         |                  |  |  |
| How to play planning poker                                                 |            |       |  |         |                  |  |  |
|                                                                            |            |       |  |         |                  |  |  |
| 1. Everyone type their number into their value field, don't hit return yet |            |       |  |         |                  |  |  |
| 2. Someone, perhaps a product owner, count down 3.. 2.. 1..                |            |       |  |         |                  |  |  |
| 3. Then, everyone hit return to submit their value                         |            |       |  |         |                  |  |  |
|                                                                            |            |       |  |         |                  |  |  |